TECHFEST, IIT BOMBAY · ROUND 1 SUBMISSION

Look-Ahead Resource Allocation for Emergency Departments Under Shifting Demand: A Multi-Agent Decision-Support Prototype

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1 Problem & Theme Chosen

We chose **"Splitting a Shared Resource When Demand Keeps Changing,"** studied in a hospital emergency department (ED). Patients of different urgency compete for limited general, oxygen-supported and ICU beds, ventilators and staff while arrivals surge unpredictably. The system is a decision-support prototype using synthetic data; clinicians retain every real decision and we claim no clinical validity.

2 Understanding of the Challenge

A fixed rule is tuned for yesterday's demand: giving the last ICU bed to the first eligible patient leaves none when sicker patients arrive minutes later. Allocation must weigh *future* demand, not only the queue. Hinson et al. (JAMA Intern Med 2024; 12,147 chest-pain visits, 3 EDs) and a scoping review of AI in ED triage (PubMed 39127019) treat AI urgency scoring as decision support and warn of overconfident predictions and poor transfer between hospitals. A waiting-time study on two Italian hospitals (40,504 and 60,230 cases) reports that queue-state features and ensemble models cut forecast error by 23–28%. Because an empty bed may lack oxygen, monitoring or staff, we allocate resource *bundles*, not bed counts.

3 Proposed Approach

A. System modules (Figure 1)

Module	Function
Patient Data Collection	Gathers patient details, symptoms, vital signs and history.
AI Triage Engine	Analyses the information and suggests an urgency category; starts as a transparent rule-based score, upgradeable to a small ML classifier.
Prioritization Engine	Orders the queue by urgency.
Resource Allocation Engine	Finds suitable beds, ICU capacity, staff and equipment.
Database	Stores patient records, triage output and resource availability.
Dashboard	Shows patient queues, available resources and shortage alerts.
Staff Approval	Lets clinicians review, confirm or override every decision; overrides are logged.

The design is two cooperating agents (triage, allocation). If independent actors are required, ED, ward and ICU units become separate agents requesting from the shared pool.

B. Allocation methodology

Baseline: a rule-based allocator assigns patients by urgency, then arrival time, to the first compatible free bed.
Proposed allocator: receding-horizon planning (MPC). At each step it:

1. estimates the arrival rate from a moving window and flags shocks with a change-detection test such as CUSUM;
2. simulates the next H hours for several candidate policies (for example, different ICU/oxygen reserve levels);
3. scores each by urgency-weighted waiting time plus penalties for unserved critical patients;
4. applies only the first decision, then re-plans.

> **Hard constraints** Critical patients are never displaced by lower-urgency ones, and stale capacity data raises an alert. We prefer planning to reinforcement learning because a beginner team can debug it; PPO is the alternative.

C. Simulation environment

A custom Gymnasium-compatible simulator generates synthetic patients with time-varying Poisson arrivals, starting at 10 general, 4 oxygen and 2 ICU beds, 1 ventilator, 3 doctors and 8 nurses.

D. Supplementary Bao-inspired module (separate from the core system)

Bao (Marcus et al., SIGMOD 2021) is a bandit wrapper using a tree-convolutional network and Thompson sampling to choose query hint sets, with fallback to the native optimizer. Our module is only Bao-inspired: a query optimizer that compares query variants and selects by measured execution time. It does not aim to reproduce the network or Thompson sampling, is separate from triage and carries none of our core claims. Intended stack: HTML, CSS and JavaScript frontend; Node.js and Express.js backend; MongoDB Atlas database; hospital management dashboard. The existing code has not yet been audited.

4 Minimum Viable

Requirement 1 · Rule-based baseline	Requirement 2 · MPC allocator vs baseline
Assigns patients by urgency, then arrival time, to the first compatible free bed.	Compared with the baseline in an abrupt respiratory-demand shock.

> **Target** Lower urgency-weighted and 95th-percentile waiting time without worse critical-patient outcomes, claimed only if it holds across at least 30 seeds with confidence intervals.

> **Fallback** A one-step lookahead reserving an ICU/oxygen margin from forecast demand.

5 Stretch Ambitions

1. One run with both an abrupt shock and gradual drift, testing reaction to a jump without over-reacting to noise.
2. Long-horizon evaluation (simulated weeks) to expose behaviour that short episodes hide.

6 Novelty/Differentiation

We claim **no new algorithm**; receding-horizon planning and ED queue forecasting are established. Our contribution is an **application-specific integration**: bundle-aware allocation with explicit look-ahead, tested under shock and drift in one long run against a fair baseline. The cited sources study triage and waiting-time prediction separately; the combined workflow is our proposed extension.

7 Evaluation Plan

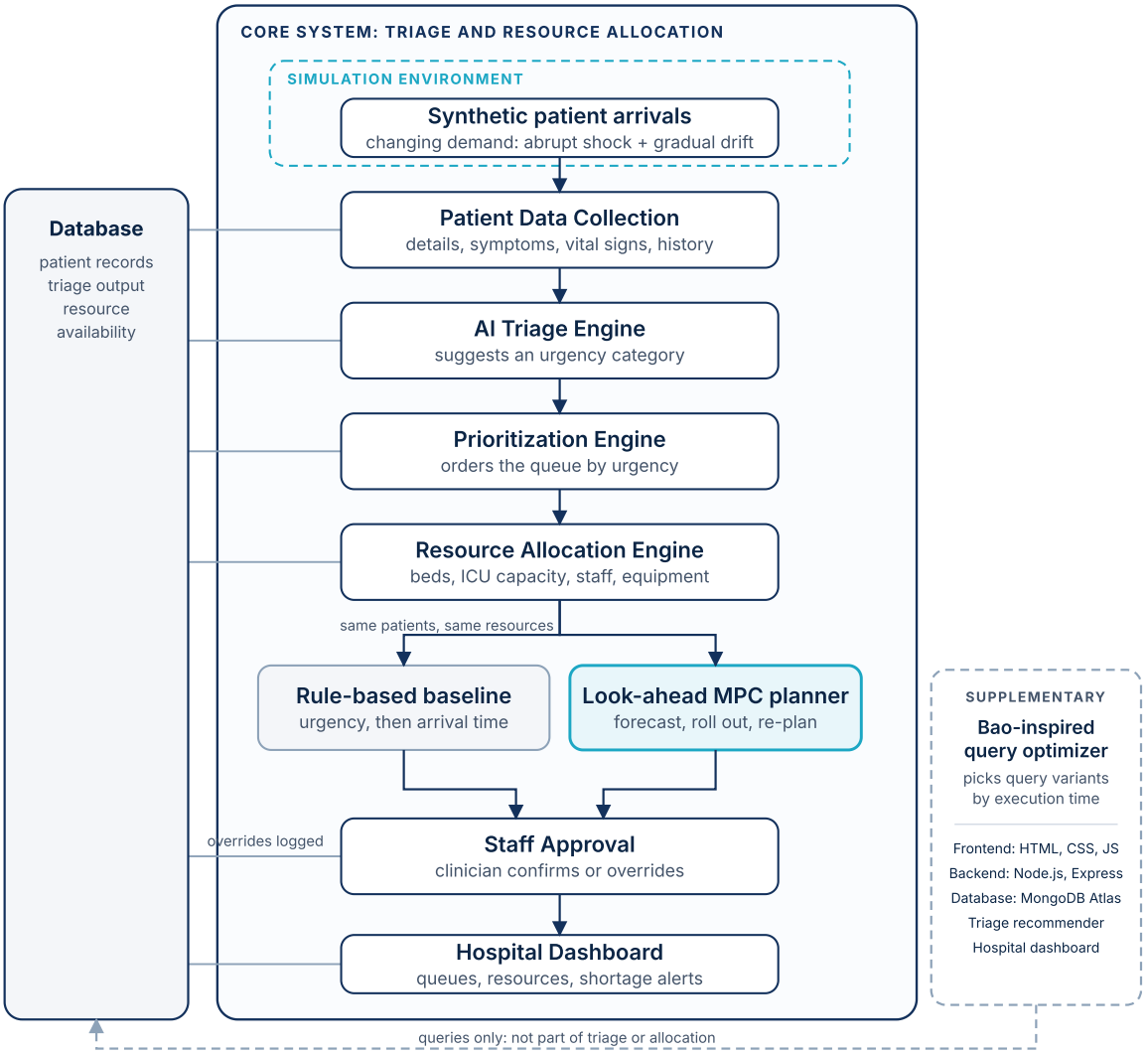
Stationary	Abrupt shock	Gradual drift	Combined
Scenario 1	Scenario 2	Scenario 3	Shock + drift in one run

Waiting	Resources and correctness	Dynamics and cost
• Mean waiting time • 95th-percentile waiting time • Urgency-weighted waiting time • Time to appropriate bed	• ICU/oxygen utilisation • Invalid assignments	• Deterioration while waiting • Shock recovery time • Compute time

> **Safety guard** An average improvement is rejected if critical-patient waiting worsens. Results will be simulation outcomes; none exist yet.

Sources cited: Hinson et al., JAMA Intern Med 2024;184(9):1125–1127, doi:10.1001/jamainternmed.2024.3219 · Scoping review of AI in ED triage, PubMed 39127019 · ED waiting-time prediction study, two Italian hospitals (full citation to be completed from the original) · Marcus et al., Bao: Making Learned Query Optimization Practical, SIGMOD 2021.

Figure 1 Proposed system architecture



Proposed design, not existing code. The rule-based baseline and the look-ahead MPC planner receive the same simulated patients and resources, so the comparison is fair. Thin grey lines show database reads and writes. The dashed Bao-inspired box is a separate, supplementary layer; its stack is the intended one and has not been audited.

8 Feasibility and Risks

Team (proposed roles, to be adjusted to actual experience):

Member	Skills	Proposed responsibilities
Aryan Jadhav	C++, basic database concepts	Integrates the dashboard with the database and builds the Bao-inspired module.
Atharva Falle	Programming logic, algorithm design	Builds the baseline allocator, constraints and planning-agent design.
Pratish	Research, data handling, documentation	Reviews literature, designs synthetic scenarios and runs evaluation.

All are beginners, so we build incrementally: **baseline first, then the planning agent.**

Risks	Mitigations
• Simulator setup time • Misleading synthetic data • An MPC that fails to beat a strong baseline	• Staged build • The one-step fallback • Honest reporting • Optional Bao module